SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 3 – Comparative Analysis

Course Code:	CSA06	**Course Title:**	Design and Analysis of Algorithms
CO Assessed:	**CO4:** Articulation (BL4, BL5)	**Assessment:**	Assessment Tool 3 – Comparative Analysis
Weightage:	10% of CIA	**Total Marks:**	100
CO4 Statement:	Solve optimization problems using dynamic programming and greedy strategies.		
Student Name:	**Beffy A Shanika**	**Reg. No.:**	**192511387**
Section / Batch:	**2025**	**Date:**	**22/08/26**

Instructions to Students

- Answer the following question. Total Marks: 100.
- Analyze the given questions thoroughly before answering.
- Compare multiple aspects such as methods, results, advantages, and limitations clearly.
- Critically evaluate the strengths, weaknesses, and implications with proper justification.
- Present meaningful insights, interpretations, and well-supported conclusions from the comparisons.

Question Type	Focus Area	Questions
Comparative Analysis	Comparative analysis of Dynamic Programming and Greedy algorithms for optimization problems.	Q1

SECTION A – Comparative Analysis

Q1 Huffman Coding vs Optimal BST (Tree Optimization)

100 marks

Compare Huffman coding and Optimal BST construction. Analyze differences in constraints and objective functions. Evaluate computational complexity. Generate insights on greedy vs DP tree design.

Code of Conduct

I Beffy A Shanika(192511387) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: beffy

RUBRICS FOR CALCULATION

Comparative Analysis					
Criteria	**Weightage**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**
Selection of Studies/Parameters	20%	Selects highly relevant and diverse studies with well-defined comparison parameters	Selects relevant studies with minor gaps in parameter definition	Limited or partially relevant selection	Poor or inappropriate selection of studies
Comparison & Differentiation	25%	Clearly compares multiple aspects (methods, results, limitations) with strong differentiation	Good comparison with minor gaps	Basic comparison with limited depth	No clear comparison; mostly descriptive
Critical Evaluation	25%	Critically evaluates strengths, weaknesses, and implications with strong justification	Provides evaluation with some justification	Limited evaluation; lacks depth	No critical evaluation provided
Synthesis & Insight Generation	20%	Generates meaningful insights and conclusions from comparisons	Some insights derived from comparison	Limited or unclear insights	No insights generated
Clarity	10%	Information is clearly presented (tables/figures) with logical structure	Generally clear with minor issues	Presentation lacks clarity or structure	Poor presentation and difficult to understand

Marks Evaluation:

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Selection of Studies/Parameters	20%				
Comparison & Differentiation	25%				
Critical Evaluation	25%				
Synthesis & Insight Generation	20%				
Clarity	10%				
Total Marks (100):					

Faculty In-charge	Course Coordinator	Head of Department
Name:	Name:	Name:
Signature: ________________	Signature: ________________	Signature: ________________
Date: ____________________	Date: ____________________	Date: ____________________

ANSWER:

CODE:

```python
import heapq
def huffman(freq):
    heap = freq[:]
    heapq.heapify(heap)
    cost = 0
    while len(heap) > 1:
        a = heapq.heappop(heap)
        b = heapq.heappop(heap)
        s = a + b
        cost += s
        heapq.heappush(heap, s)
    return cost
def optimal_bst(freq):
    n = len(freq)
    dp = [[0] * n for _ in range(n)]
    for i in range(n):
        dp[i][i] = freq[i]
    for length in range(2, n + 1):
        for i in range(n - length + 1):
            j = i + length - 1
            total = sum(freq[i:j + 1])
            dp[i][j] = float('inf')
            for r in range(i, j + 1):
                left = dp[i][r - 1] if r > i else 0
                right = dp[r + 1][j] if r < j else 0
                dp[i][j] = min(dp[i][j], left + right + total)
    return dp[0][n - 1]
n = int(input())
freq = list(map(int, input().split()))
print("Huffman Cost:", huffman(freq))
print("Optimal BST Cost:", optimal_bst(freq))
```

INPUT:

```
5
10 20 30 40 50
```

OUTPUT:

```
Huffman Cost: 330
Optimal BST Cost: 300
```